

Topic: **Recursion, Dictionaries**

Date: **2026/09/20**

① Check My Understanding

Questions completed: **5 / 5**

Answers revised after AI hints: **0 / 5**

② My Misconception

Before: I thought...

None. I did not identify or correct any misconception during the five questions. I was able to answer all five questions smoothly.

Now: I understand...

I understand the concepts of recursion and dictionaries covered in today's lecture, including recursive termination, separate scopes for recursive calls, dictionary key/value properties, memoization, and mutation of dictionaries passed to functions.

③ Challenge the AI

One AI-generated question I challenged:

Question 2: "When a recursive function calls itself, the new recursive call uses the same local variables as the previous call..."

Why?

- ☐ Ambiguous
- ☐ Oversimplified
- ☐ Technically questionable
- ☒ **Too easy**
- ☐ Other: _____

Brief explanation:

The question was too easy because it directly tested the fact that each recursive call creates its own function scope, which was explicitly covered in the lecture. The AI also said the question would be slightly harder, so I expected it to require more conceptual reasoning.

④ One-Minute Reflection

One thing I am still unsure about:

I am still thinking about the relationship between tuple immutability and dictionary keys. In particular, I want to understand more clearly what happens when a tuple contains a mutable object, and how that relates to whether the tuple can be used as a dictionary key.